Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/03/2025
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Brain-Related Assessments for Investigating the Neurophysiology of SANS (BRAIN-SANS)		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:	Spaceflight Associated Neuro-ocular Syndrome (SANS) remains an important and unmitigated risk to long-duration spaceflight. Current hypotheses suggest that the lack of gravity leads to fluid shifting towards the head, resulting in congestion and/or elevated pressures in the cranial, vascular, and/or lymphatic compartments. NASA is conducting 30-day head-down tilt (HDT) experiments to test SANS countermeasures at the :envihab facility in Cologne, Germany. We propose to provide numerous key measurements in support of these planned 30-day missions. We will focus in particular on providing a toolkit for detailed neurophysiological and fluid shift assessment and monitoring suitable for measuring both SANS- and countermeasure-related changes. These tools will be designed to complement the standard ocular measures used for SANS diagnosis and monitoring (e.g., optical coherence tomography (OCT), ocular ultrasound (US), fundoscopy, visual acuity). Our proposed measures will include: • Relative intracranial pressure (ICP) measurements via distortion product otoacoustic emissions (distortion product otoacoustic emissions: DPOAE). • Blood volume shifts along the body axis via near-infrared spectroscopy (NIRS). • Intracranial blood inflow and outflow, via internal jugular vein (IJV) and carotid artery (CA) ultrasound cross-sectional imaging and Doppler. • Cerebral pulsatility assessment, per our parabolic flight and SPACE-COT (Studying Physiological and Anatomical Cerebral Effects of CO2 and Tilt) :envihab NIRS study. • Blood pressure at the level of the head via local, cuffless superficial temporal artery tonometry. • Sagittal sinus blood volume imaging and monitoring using diffuse optical tomography (DOT). • Cerebral edema assessment based on H2O concentration imaging, similar to that used in previous altitude sickness studies. • Cerebral electrical activity, via electroencephalogram (EEG) measurements. • Dynamic cerebral autoregulation (CAR) assessment during countermeasure (CM) challenges, which can be derived from the NIRS signals used in the above measurements. Our tools will be made fully compatible with the planned SANS countermeasures, as well as with those of other teams proposing specific CMs. Along with the measures, we will provide the necessary expertise and analysis to quantify physiological changes associated with SANS countermeasures deployed during the 30-day HDT campaigns at :envihab. Our specific aims are as follows: Aim 1: Develop an integrated collection of hardware to support multiple simultaneous, continuous brain monitoring/imaging capabilities, and ensure the hardware and measurements are fully compatible with all countermeasures deployed during the :envihab missions. Aim 2: Characterize and quantify individual subjects' physiological responses to each planned condition, including comparative assessment of SANS countermeasures. Aim 3: Relate neurophysiological changes over the 30-day HDT—both with and without SANS-CMs—to cognitive and operational performance, sleep, mood, and ocular measures. This will include the Cognition battery, psychological/mood surveys, and a suite of ocular measures (OCT, fundoscopy). We will obtain as many measures as possible through data sharing and investigate the relationship of our neurophysiological measures to each of these outcome assessments. Jointly, the planned measures and Aims will enable NASA to quantitatively evaluate and compare the (neuro)physiological changes and fluid shifts associated with HDT and SANS countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will involve developing a detailed toolbox of measures for assessing brain physiology. These technologies will be compatible with the planned SANS countermeasures, and hence could be deployed in multiple other settings on Earth, ranging from intensive care units to exercise settings. The detailed and simultaneous monitoring of numerous cerebral physiology variables is expected to provide new insights into how the brain responds to various types of interventions. As such, the data could provide insights into how the body--and brain in particular--responds to pre-syncope, bedrest, exercise, fluid shifts, and sequestration of blood in the extremities. All of these have implications in medicine here on Earth.
	Background Spaceflight associated neuro-ocular syndrome (SANS) is an unsolved risk for astronauts on long-duration missions. When diagnosed from Frisen grade papilledema on fundoscopy, some 10 of 68 astronauts have exhibited SANS, although related ocular findings are more common (e.g., acquired hyperopia, globe flattening, choroidal folds, retinal fiber nerve layer thickening), and current estimates are closer to a 75% prevalence of SANS in astronauts on 6-month missions. Unexpectedly, SANS signs do not always spontaneously resolve upon return to Earth gravity. While the cause of SANS is unknown, the hyperopia, globe flattening, and choroidal folds—coupled with typically normal or slightly elevated intraocular pressure (IOP)—suggests that intracranial pressure (ICP) may be elevated as compared to average Earth levels. Various pathophysiological mechanisms have been proposed for SANS, with particular suspicions regarding cephalad fluid shifts. SANS Countermeasures Most hypotheses regarding SANS involve headward fluid shifts as a factor, and various proposed SANS countermeasures (CMs)—including lower-body negative pressure (LBNP), veno-constrictive thigh cuffs (VTC), inspiratory resistance threshold devices (ITD), and artificial gravity (AG)—all involve "mechanical" redistribution of body fluids away from the head. Understanding the relative benefits of such CMs calls for assessments of perfusion and fluid flow into, within, and out of the cranium not only for potentially assessing and monitoring SANS but also to help quantify and compare the effect sizes of various CMs. SANS-CM Study at DLR's Envihab Facility

Task Progress:	To address the lack of SANS CMs, NASA negotiated a plan with the German Aerospace Center's :envihab facility to conduct 30-day head-down tilt (HDT) bedrest studies—the SANS-CM study. This effort currently includes 4 study arms: 1. 6o HDT bedrest alone (Reference) 2. 6o HDT bedrest plus two 3-hour periods per day seated upright (Seated CM) 3. 6o HDT bedrest plus two 3-hour periods per day of LBNP (LBNP CM) 4. 6o HDT bedrest plus one ~1-hour period of exercise followed by 6 hours of VTC (Exercise CM) Each arm targeted n=12 subjects, and different investigators were involved in different aspects of the overall SANS-CM study. Our contribution was the BRAIN-SANS (sub)project. BRAIN-SANS Contribution This BRAIN-SANS project (a component of SANS-CM) seeks to provide a wide range of brain-related measures for all subjects in all study arms. These include changes in (i) intracranial pressure (ICP), (ii) blood flow in/out of the brain, (iii) cerebral blood flow velocity, (iv) brain perfusion and oxygenation, (v) blood distribution along the body axis, (vi) intracranial pulsatility, (vii) sagittal sinus imaging of potential venous congestion, (viii) intracranial water concentration, (ix) functional brain activation, (x) electrical brain activity, as well as (xi) cognitive performance data (Cognition). We also plan to compare these measures with measures from other groups including ocular measures, mood and sleep, and 1-carbon single nucleotide polymorphisms. ACHIEVEMENTS IN YEAR 5 Activities that were completed or will be completed by the end of the 5th year of the project are as follows: Summary of All BRAIN-SANS Data: Overall, data collection went very well, with 6,615 of 6,780 expected data files—or 97.6% of all data files—accounted for. The minor amount of missing data arose from one subject dropout, plus occasional device faults or running behind schedule which prevented full data completion. Ongoing Data Analysis: The majority of year 5 involved data quality assurance, preprocessing and analysis. Datasets being processed include distortion product otoacoustic emissions (DPOAE) (Dr. Voss; manuscript in preparation); IJV/ICA ultrasound imaging/flow data (Dr. Bershad; statistical analysis partially complete); NIRS fluid shift data (Dr. Strangman; presented at IWS2024 and manuscript in preparation); cerebrovascular pulsatility (Dr. McCaffrey; being presented at IWS2025 and manuscript in preparation); sagittal sinus congestion (Dr. Joshi; imaging analysis underway); cerebral oxygenation and cerebral water/CSF (Dr. Forselius; preprocessing underway); Cognition data (Dr. Basner; analysis nearing completion). In addition, data sharing with Drs. Huang, Zwart, and Clement were initiated, involving data transfer, dataset harmonization, and discussion of primary and secondary variables of interest. Analysis and manuscript preparation on all of the above datasets will continue in the coming year. Preliminary Results: Cerebral blood volume showed a small shift into the head in the Control condition coupled with a large shift away from the legs. Transitioning to Seated led to no change in blood in the head or chest, substantial increases in the thigh, and large increases in the calf. The LBNP condition significantly reduced the blood volume in the chest and significantly increased blood volume in both the thigh and calf, to equal extents. Finally, the Exercise+VTC condition led to no significant changes in chest blood volume but greatly increased the blood in both the thigh and calf. This was sustained through the exercise period, the 30-minute "rest/gap" (when some loss of blood in the legs was anticipated), and increased further when the VTC was tightened. The effect size for Exercise+VTC was larger than seated, which was unexpected. Initial findings for DPOAE, IJV/CSA, as well as cerebrovascular pulsatility, also show distinct changes associated with the various CMs, although the relative effect sizes appear to be ranked differently across CMs depending on the measures. We will finalize individual analyses first before engaging in detailed inter-comparisons of the various neurophysiological measures collected via BRAIN-SANS. The remainder of year 5 and the coming year will involve further analysis and manuscript preparation and submission for all of the collected datasets. SUMMARY Year 5 provided major progress on multiple datasets within the BRAIN-SANS data collection suite. The very low rate of data loss and high-quality data (except from periods of uncontrolled subject motion during our multi-hour recordings) provide a strong base from which to address numerous hypotheses about the neurophysiology in the HDBR SANS analog as well as the effects of the investigated SANS countermeasures.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
Abstracts for Journals and Proceedings	Thoolen S, Zhang Q, Ivkovic V, Voss S, Moestl S, Frett T, Tank J, Wu J, Bershad EM, Strangman GE. "Brain-sans: Brain-related assessments for investigating the neurophysiology of sans – 2024 update." 2024 NASA Human Research Program Investigators' Workshop, Galveston Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston Texas, Feb 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Voss SE, Thoolen S, Moestl S, Frett T, Strangman GE. "Brain-sans: What do DPOAEs tell us about intracranial pressure changes during experiments designed to understand sans?" 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024